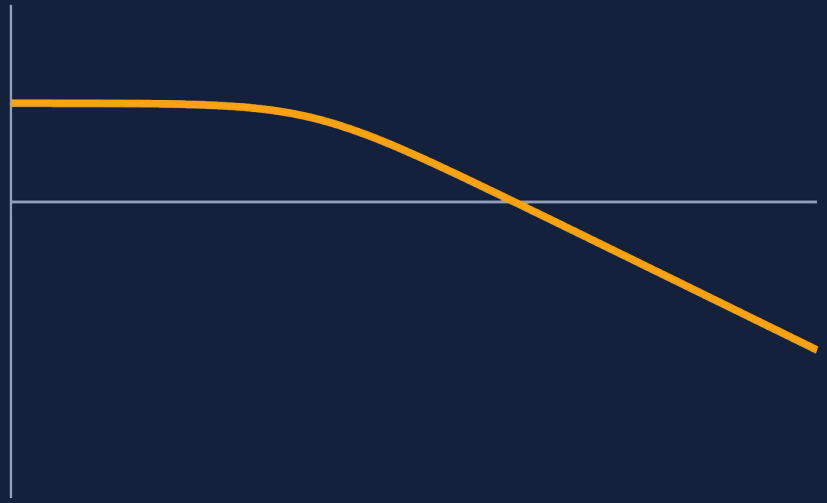
# Frequency-Domain Specifications



Control Systems (22EEG53)

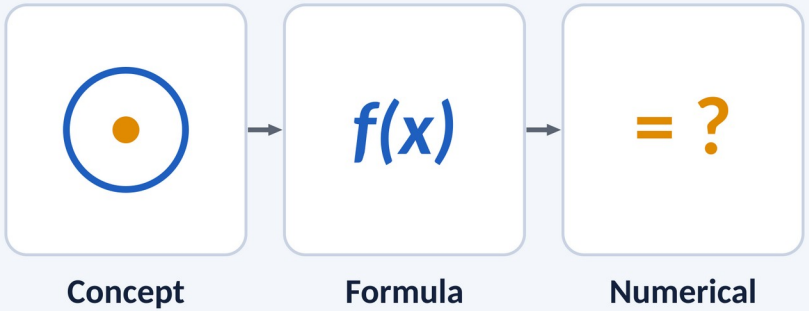
Unit 3: Frequency Domain Analysis

Topic: Peak resonance, resonant frequency, bandwidth, cut-off rate; correlation with time response

**Prof. Mitesh Astik**

# Learning Outcomes

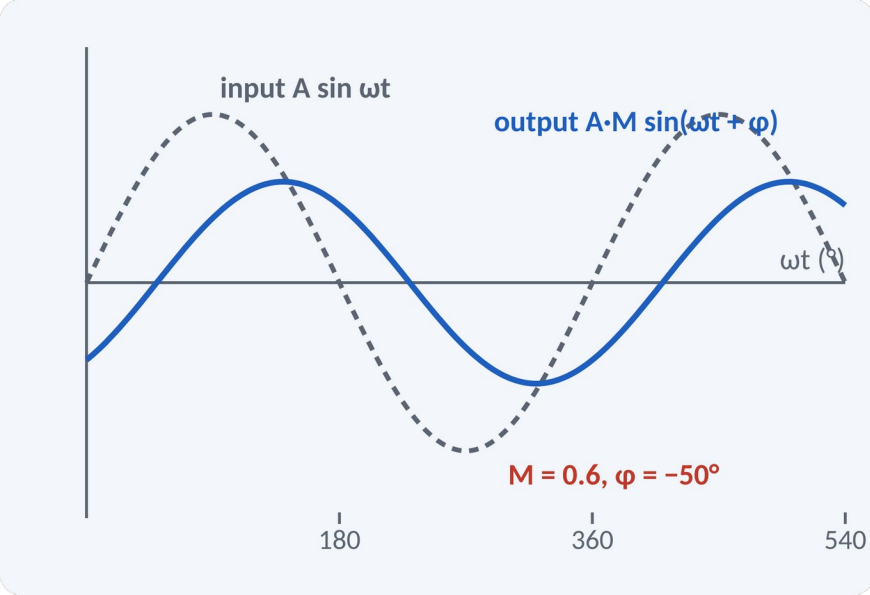
- Define Mr, ωr, ωb and cut-off rate.
- Derive Mr, ωr and ωb for a second-order system.
- Relate frequency and time-domain specifications.
- Find ζ and ωn from frequency-domain data.




**Linked to CO3**
Analyse the performance indices of a system in the frequency domain.

# Recap: Sinusoidal Steady State

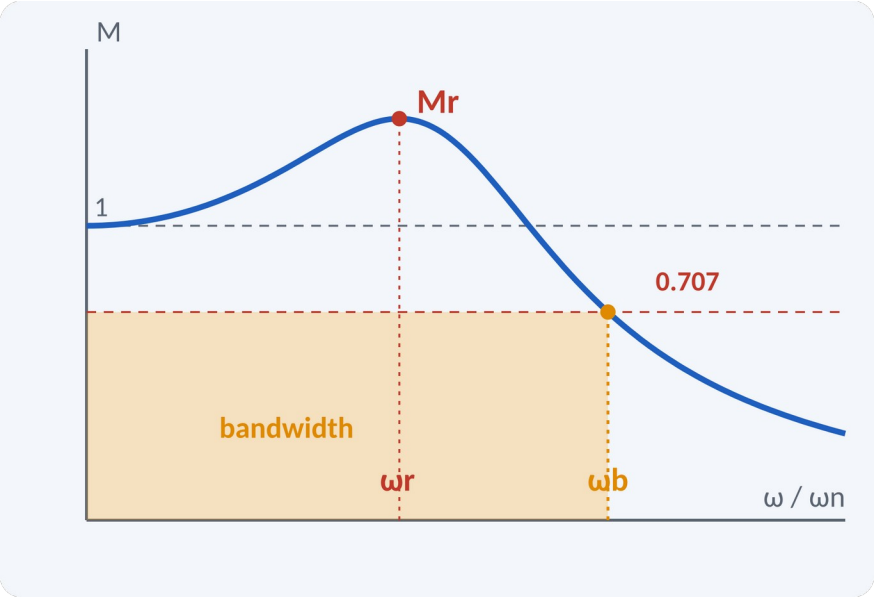
- Apply r(t) = A sin ωt to a stable system.
- Steady-state output is also sinusoidal.
- Amplitude is scaled by M = |G(jω)|.
- Phase is shifted by φ = ∠G(jω).




**$c(t) = A \cdot M(\omega) \cdot \sin(\omega t + \varphi(\omega)), \quad M = |T(j\omega)|, \quad \varphi = \angle T(j\omega)$**
T(s) closed-loop transfer function, ω in rad/s

# Motivation: Why Frequency Domain?

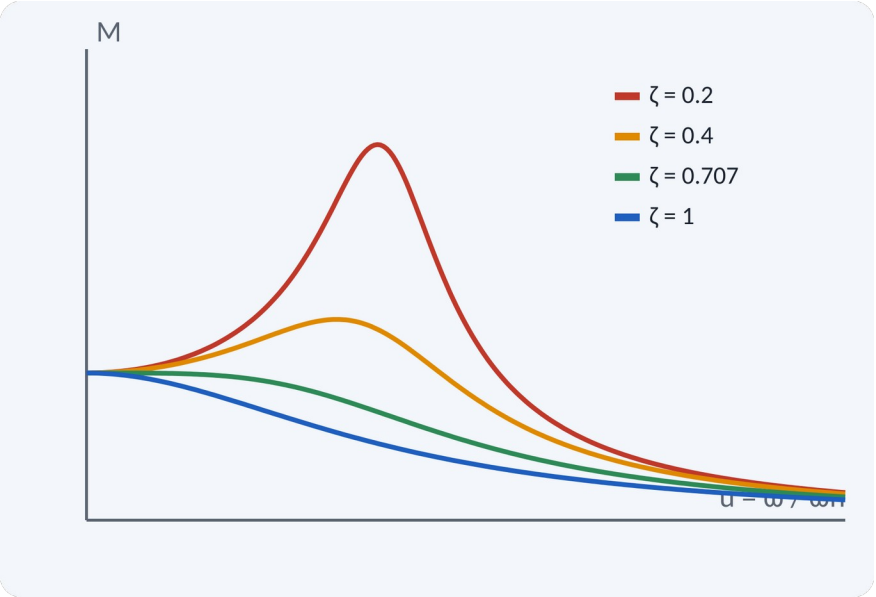
- Audio amplifier: which frequencies pass?
- Servo: how fast a command can it follow?
- Measure response with a signal generator.
- Design without solving differential equations.




**Bandwidth indicates speed; resonant peak indicates damping**

# Frequency Response of a Second-Order System

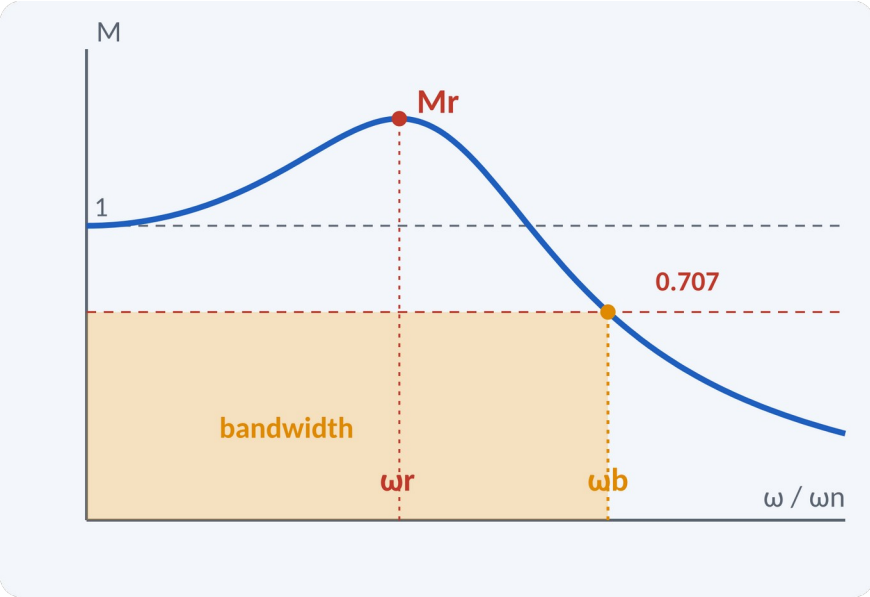
- Closed-loop $T(s) = \omega_n^2/(s^2 + 2\zeta\omega_n s + \omega_n^2)$.
- Put $s = j\omega$ and $u = \omega/\omega_n$.
- Magnitude depends on u and ζ only.
- Low ζ gives a sharp peak near u = 1.




**$M(u) = 1/\sqrt{[(1 - u^2)^2 + (2\zeta u)^2]}$;  $\varphi(u) = -\tan^{-1}[2\zeta u/(1 - u^2)]$**
u = ω/ωn normalised frequency

# Frequency-Domain Specifications

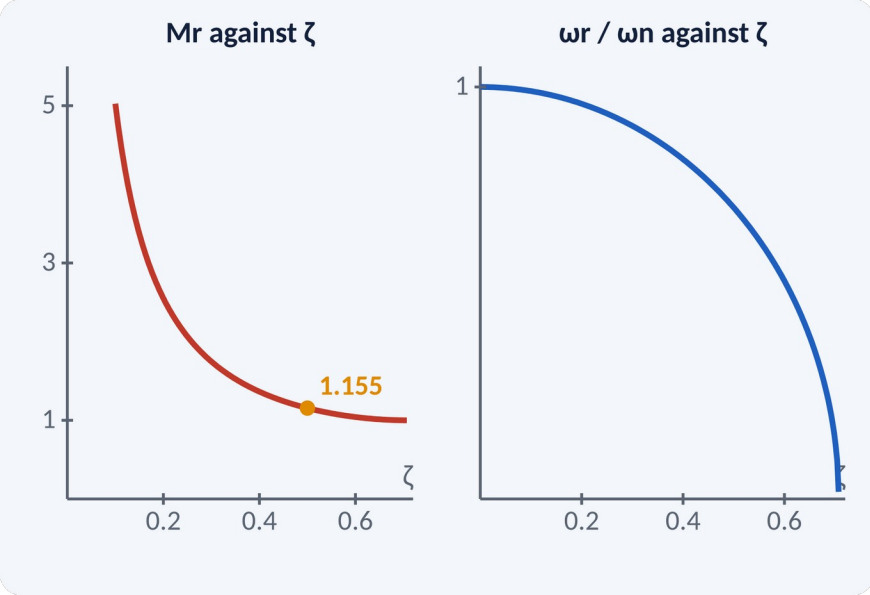
- Resonant peak Mr: maximum value of M.
- Resonant frequency ωr: frequency at the peak.
- Bandwidth ωb: where M falls to 0.707 (−3 dB).
- Cut-off rate: slope of M at high frequency.
- Gain margin, phase margin: next lecture.




**Mr = Mmax/M(0);  ωb: M(ωb) = 0.707·M(0)**
Mr dimensionless (or dB), ωr and ωb in rad/s

# Resonant Peak and Frequency

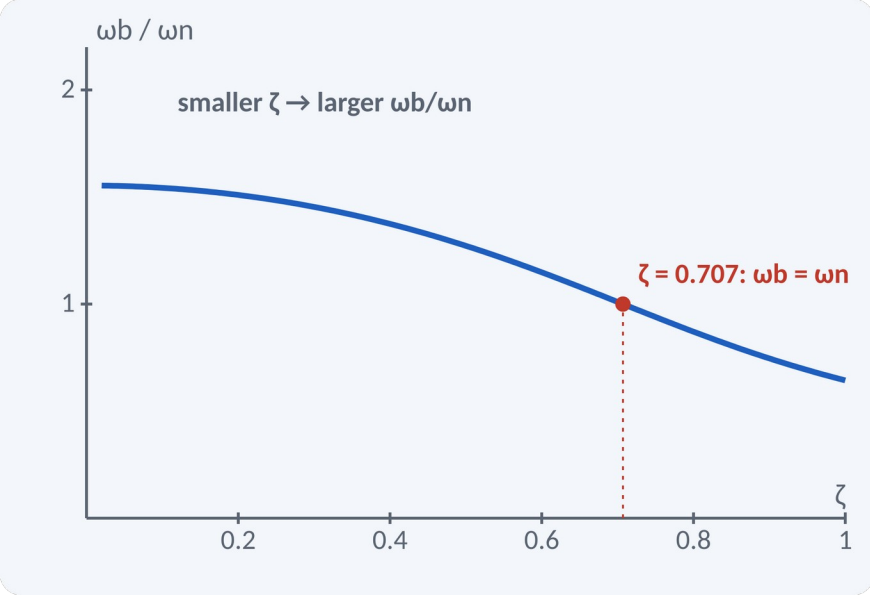
- Set dM/du = 0 to find the peak.
- A peak exists only for ζ < 0.707.
- Mr depends only on ζ.
- ωr is always less than ωn.




**$\omega_r = \omega_n\sqrt{(1 - 2\zeta^2)}$;  $M_r = 1/(2\zeta\sqrt{(1 - \zeta^2)})$  $(0 < \zeta < 0.707)$**
For ζ ≥ 0.707: no peak, Mmax = 1 at ω = 0

# Bandwidth

- Solve M(u) = 0.707 for u.
- ωb/ωn depends only on ζ.
- Larger bandwidth gives faster rise time.
- For ζ = 0.707, ωb = ωn.




**ωb = ωn·[(1 − 2ζ²) + √(4ζ⁴ − 4ζ² + 2)]^½**
ωb in rad/s

# Cut-off Rate

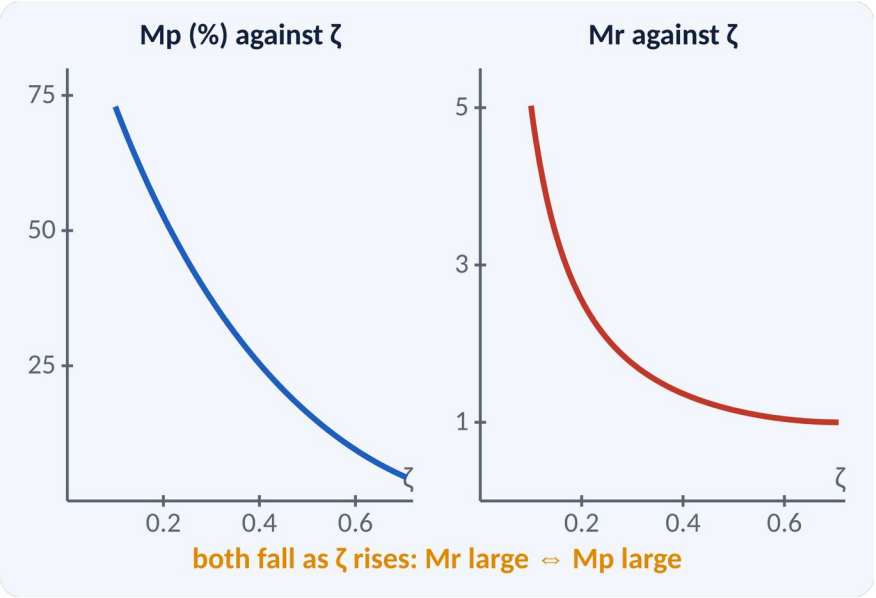
- Slope of the log-magnitude at high frequency.
- Second order: −40 dB/decade.
- First order: −20 dB/decade.
- Steeper slope rejects noise better.




**At high ω: $|T| \approx (\omega n/\omega)^2 \rightarrow$ slope = −40 dB/decade**
dB = 20 $\log_{10}$ M

# Correlation: Time and Frequency

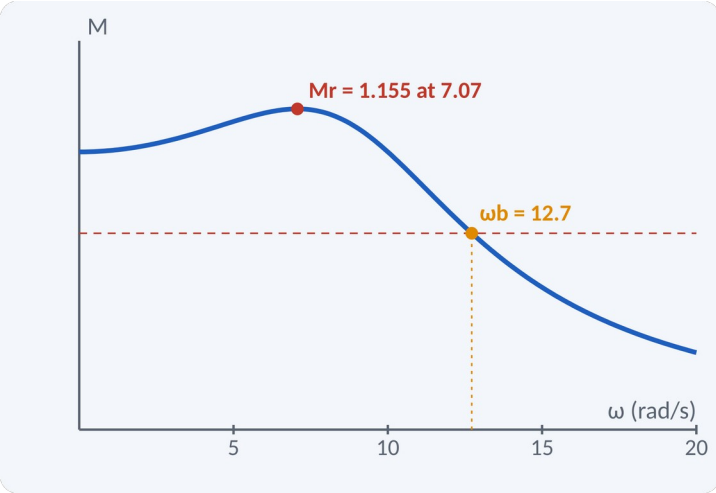
- Both Mp and Mr depend on ζ only.
- Large Mr means large overshoot.
- Bandwidth is inversely related to rise time.
- High ωb gives short tr and ts.
- ωd and ωr both scale with ωn.




**Mr ↑ ⇔ Mp ↑ ; ωb ↑ ⇔ tr ↓ ; tr·ωb ≈ constant for fixed ζ**

# Solved Example 1: Find Mr, ωr, ωb

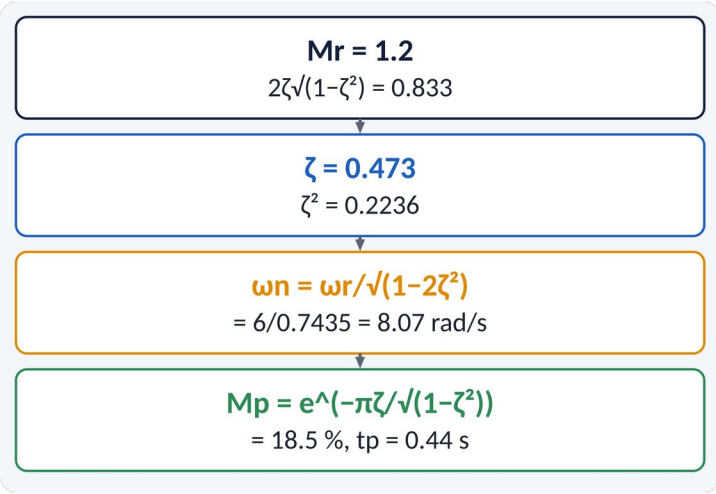
- A system has ζ = 0.5 and ωn = 10 rad/s. Find Mr, ωr, ωb.
- 1. ζ < 0.707, so a peak exists.
- 2. Mr = 1/(2·0.5·√0.75) = 1.155 (1.25 dB).
- 3. ωr = 10√(1 − 0.5) = 7.07 rad/s.
- 4. ωb = 10[(0.5) + √(0.25 − 1 + 2)]^½ = 10 × 1.272 = 12.7 rad/s.




**Answer: Mr = 1.155, ωr = 7.07 rad/s, ωb = 12.7 rad/s**

# Solved Example 2: Specs from Data

- A servo shows Mr = 1.2 at ωr = 6 rad/s. Find ζ, ωn, and the expected Mp and tp.
- 1. $2\zeta\sqrt{(1-\zeta^2)} = 1/1.2 = 0.833$, so $\zeta^2(1-\zeta^2) = 0.1736$.
- 2. $\zeta^2 = 0.2236$ (root with $\zeta < 0.707$), so $\zeta = 0.473$.
- 3. $\omega n = \omega r/\sqrt{(1-2\zeta^2)} = 6/0.7435 = 8.07$ rad/s.
- 4. Mp = e^(−π·0.473/0.881) = 18.5 %.
- 5. ωd = 8.07 × 0.881 = 7.11, tp = π/7.11 = 0.44 s.




**Answer: ζ = 0.473, ωn = 8.07 rad/s, Mp = 18.5 %, tp = 0.44 s**

# Common Mistakes and Exam Tips

## Common mistakes

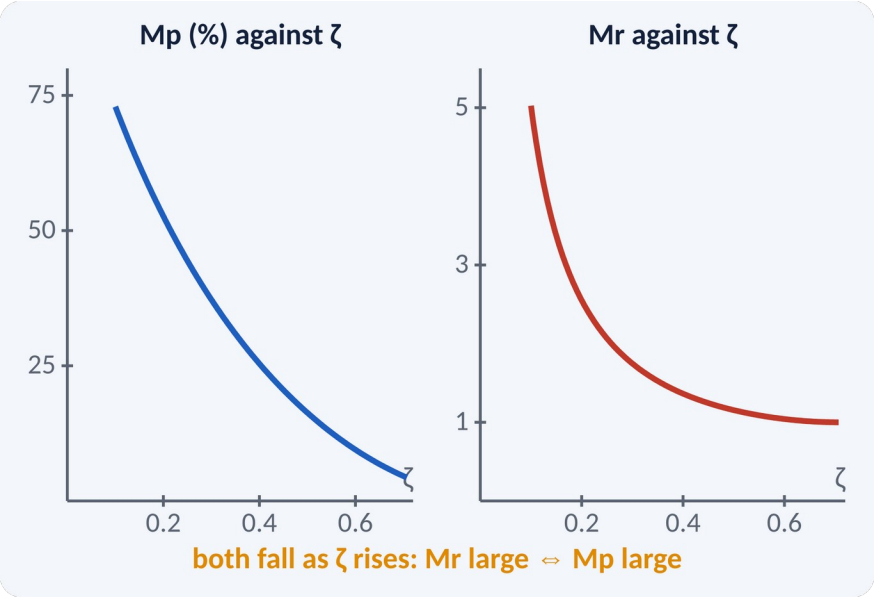
- Using the Mr formula for $\zeta \geq 0.707$.
- Taking bandwidth at the peak instead of at 0.707.
- Mixing dB and linear values of M.
- Using $\omega r$ in place of $\omega n$.
- Forgetting that the cut-off rate is per decade.

## Exam tips

- Check $\zeta < 0.707$ before using Mr and $\omega r$.
- Bandwidth: find where $M = 0.707\ M(0)$.
- Convert: $dB = 20 \log_{10} M$.
- Solve $\zeta^2$ from the quadratic and choose the small root.

# Summary

- $M = |T(j\omega)|$ and $\varphi = \angle T(j\omega)$ describe frequency response.
- $M_r = 1/(2\zeta\sqrt{(1-\zeta^2)})$ and $\omega_r = \omega_n\sqrt{(1-2\zeta^2)}$ for $\zeta < 0.707$.
- Bandwidth is the −3 dB frequency; larger means faster.
- Cut-off rate of a second-order system is −40 dB/decade.
- Mr tracks overshoot; ωb tracks rise time.




**Mr, Mp depend on ζ only; ωb, ωr scale with ωn**

# Quick Quiz

**Q1. Mr for ζ = 0.5 is about:**
(a) 0.5 (b) 1.0 (c) 1.155 (d) 2.0

**Ans: (c)**

**Q2. A resonant peak exists only when:**
(a) ζ > 1 (b) ζ < 0.707 (c) ζ = 1 (d) ζ > 0.707

**Ans: (b)**

**Q3. Bandwidth is the frequency where M falls to:**
(a) 0.5 (b) 0.707 (c) 0.9 (d) 0.1

**Ans: (b)**

**Q4. Cut-off rate of a second-order system:**
(a) −20 dB/dec (b) −40 dB/dec (c) −60 dB/dec (d) 0

**Ans: (b)**

**Q5. For ζ = 0.707 the bandwidth equals:**
(a) ωn/2 (b) ωn (c) 2ωn (d) ωr

**Ans: (b)**

# References


1. **Benjamin C. Kuo, Automatic Control Systems**
   7th Ed., PHI, 2007 (Units 3, 4, 5 textbook)

2. **P. S. Satyanarayana, Concepts of Control Systems**
   2nd Ed., Dynaram Publishers, Bangalore, 2001 (reference)

3. **K. Ogata, Modern Control Engineering**
   5th Ed., Pearson Education Asia / PHI, 2009 (reference)


**Next lecture: Bode plots, gain margin and phase margin**